3D Field of Junctions on New Volumetric Denoising Domains

Samudraneel Sarkar, Sakshi Patil*, Nikhil Kumar*, Manpreet Singh*

MNNIT Allahabad

{samudraneel.20233249, patil.20233209, nikhil.20233202, manpreet.20233180}@mnnit.ac.in

Abstract

3D Field of Junctions (FoJ) is a training-free denoiser that fits each patch of a volume with planar wedges meeting at a common vertex and averages the overlapping fits. Its demonstrated applications are low-dose X-ray CT, cryo-electron tomography, and point-cloud denoising. We apply the authors' publicly released implementation to three domains outside that set, porous-media micro-CT, an MRI phantom, and binary porous media, and compare it with Gaussian, median, and total-variation (TV) filtering under controlled noise levels. On micro-CT, classical filters are stronger at moderate noise, but FoJ overtakes all of them once the input falls below about 11 dB PSNR, exceeding the best classical result by 3.2 dB at the highest noise tested while retaining boundary structure the filters lose. The other two domains mark the boundary of the method: it does not help where the structure is smooth or smaller than the patch scale.

1 Introduction

Volumetric images acquired under dose and time limits are noisy, and denoising is a routine step in their analysis. Classical denoisers include local filters such as Gaussian smoothing and median filtering, variational methods such as total-variation (TV) minimization [1, 2], and patch-recurrence methods such as non-local means [3] and BM3D [4]; learned denoisers trained on paired data [5] dominate at moderate noise levels.

Field of Junctions (FoJ) [6] is a training-free structural prior that models each image patch as constant-intensity regions separated by boundaries meeting at a vertex, called a junction. 3D FoJ [7] extends the model to volumes: three planes meeting at a common vertex divide each cubic patch into constant-valued wedges, and overlapping patch fits are averaged into a denoised volume. The method targets the low-SNR regime and was demonstrated on low-dose X-ray CT, cryo-electron tomography, and point-cloud denoising.

We apply the released 3D FoJ implementation to three domains outside that set. Grayscale micro-CT of porous rock is a natural fit for the wedge model, since grains and pores form near-constant regions with sharp, piecewise-planar boundaries. An MRI phantom and two binary porous-media volumes are poor fits: the first has smooth tissue intensities, the second has structure smaller than the patch scale. Testing all three under controlled noise shows where the method transfers. Our implementation and experiment scripts are available in the accompanying repository.[1]

*Equal contribution; order determined by coin flip.

[1] https://github.com/samudraneel05/foj3d-factor-study

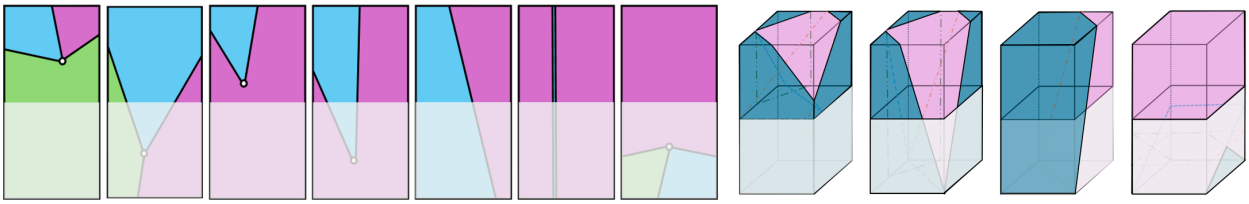


Fig. 1: Junction schematics reproduced from [7] (CC BY 4.0). Left: 2D junctions with three wedge regions. Right: 3D junctions with two regions formed by planes through a common vertex.

σ	Noisy	Gaussian	Median	TV (best w)	FoJ (best R)
15%	16.89	**28.51**	27.01	27.65	26.55
30%	12.00	25.26	22.26	**26.03**	25.64
40%	10.28	23.39	20.04	23.32	**24.93**
55%	8.76	21.05	17.60	16.30	**24.29**

Table 1: PSNR (dB) on the Bentheim volume. FoJ overtakes all classical baselines once the input PSNR falls below ∼11 dB (+1.5 dB at σ=40%, +3.2 dB at σ=55%) while trailing at moderate noise.

2 Methodology

FoJ partitions a volume into overlapping cubic patches of R voxels per side. Inside each patch, three planes through a common vertex split the patch into wedge-shaped regions, and each region takes a single constant intensity (Fig. 1). A patch is fitted by searching over plane orientations and vertex positions to minimize the squared error against the input patch; each region's intensity is the mean of the voxels it covers. Since every voxel belongs to several overlapping patches, the denoised volume is the average of all patch fits covering that voxel. The patch size R is the method's main hyperparameter: it sets the largest noise scale that averaging can absorb, and the smallest structure a wedge model can represent.

Because each region is a constant, the model removes noise by replacing every wedge of the volume with a regional mean. Two consequences follow. Structure that is not piecewise-flat at the patch scale, such as smooth texture or features smaller than R, cannot be represented and is averaged away; and the wedge approximation is a coarse model, so it pays off only when the noise it removes exceeds the bias it introduces.

We use the authors' released implementation at its default configuration.

3 Results and Discussion

Setup. We test three domains with public ground truth: a 192^3 crop of Bentheim sandstone micro-CT (5.85 μm voxels, grains tens of voxels across [8]) under additive Gaussian noise at $\sigma \in \{15\%, 30\%, 40\%, 55\%\}$ of the dynamic range; the BrainWeb T1 phantom (181×217×181 [9]) under Rician noise at $\sigma \in \{3\% \ldots 13\%\}$; and two binary porous-media volumes (a synthetic sphere pack and a segmented Berea scan [10]) with feature sizes of about 2 voxels, under Gaussian noise at σ=15%. Baselines are Gaussian filtering, a 3^3 median filter, and TV [1, 2] at three weights, all under identical conditions; FoJ runs at $R \in \{6, 8, 12\}$, stride 2. We also report a boundary F-score over gradient-magnitude edge voxels, since edge preservation is FoJ's distinguishing feature.

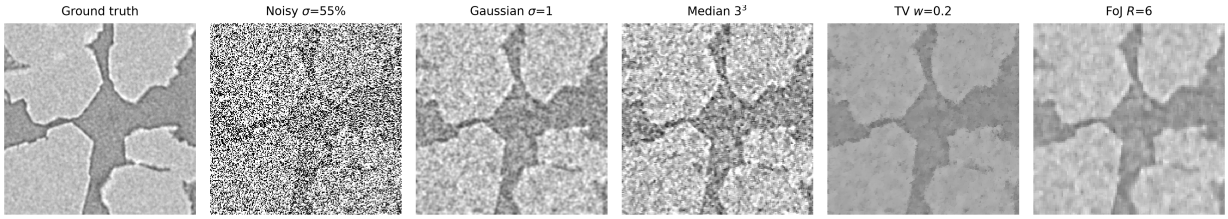


Fig. 2: Bentheim micro-CT, mid-slice, σ=55% Gaussian noise. FoJ (R=6) preserves the sharp grain–pore interfaces that Gaussian, median, and TV filtering blur.

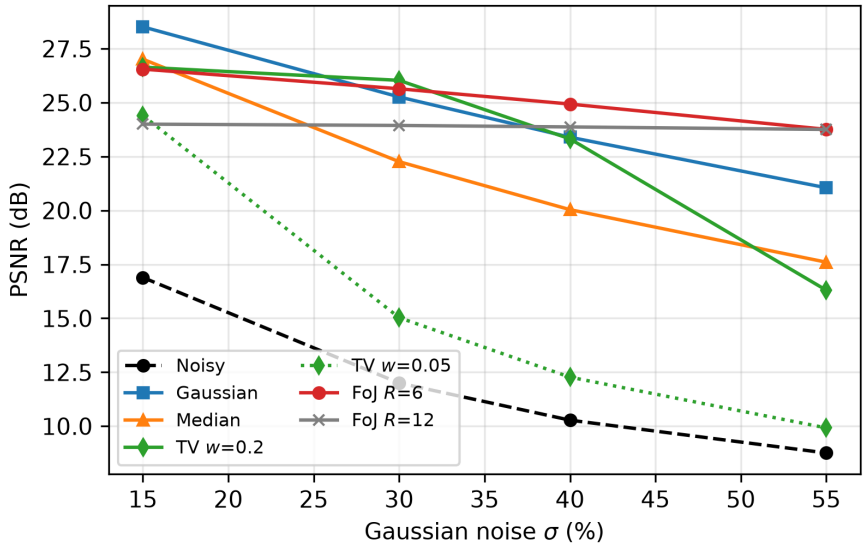


Fig. 3: PSNR versus noise level on Bentheim micro-CT. FoJ (red) overtakes the classical filters between σ=30% and 40%.

Porous-media micro-CT. Table 1 and Fig. 3 show a clean crossover. FoJ trails at $\sigma \leq 30\%$, overtakes every classical baseline between 30% and 40%, and its margin grows with noise. The advantage is concentrated at boundaries: at σ=55% FoJ retains boundary-F1 = 0.85–0.88 against 0.54 for the best TV weight, and Fig. 2 shows the grain–pore interfaces preserved where TV loses them. The best patch size grows with noise: R=8 wins at σ=55%.

The other two domains. On the BrainWeb T1 phantom [9] under Rician noise, FoJ trails a median filter by 1–5 dB at every tested level: smooth tissue intensities are not piecewise-flat, so the wedge model's bias outweighs its denoising. On two binary porous-media volumes (a synthetic sphere pack and a segmented Berea scan [10]) whose features span about 2 voxels, FoJ's output collapses toward the mean (Fig. 4), reaching 7.1 dB on a 19.5 dB input: structures smaller than the patch cannot be represented by wedges.

Fig. 5 places all results on one axis: FoJ's margin over the best classical filter versus input PSNR. On compatible structure the margin grows with noise and crosses zero near an input PSNR of 11 dB; the control domains stay negative because their violation is structural rather than noise-driven.

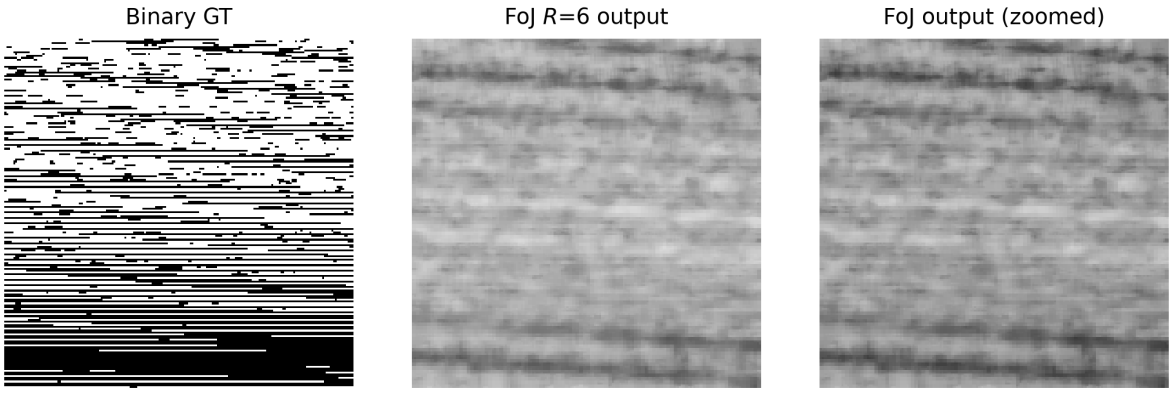


Fig. 4: Collapse on sub-patch-scale binary structure: features of about 2 voxels cannot be represented inside R=6 patches and the output falls toward the mean.

4 Conclusion

Across three volumetric domains outside the original study's set, 3D FoJ succeeds on the one whose structure its wedge model can express at high noise: porous-media micro-CT, where it outperforms Gaussian, median, and TV denoising once the input falls below about 11 dB PSNR, with a +3.2 dB margin and sharply better boundary preservation at the highest noise tested. Smooth MRI tissue and sub-patch-scale binary volumes fail. The released implementation is an effective training-free denoiser for low-SNR volumes with piecewise-flat structure; outside that structure, classical filters remain the better choice.

References

[1] L. I. Rudin, S. Osher, and E. Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D*, 60(1–4):259–268, 1992.

[2] A. Chambolle. An algorithm for total variation minimization and applications. *Journal of Mathematical Imaging and Vision*, 20(1):89–97, 2004.

[3] A. Buades, B. Coll, and J.-M. Morel. A non-local algorithm for image denoising. In *CVPR*, 2005.

[4] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian. Image denoising by sparse 3-D transform-domain collaborative filtering. *IEEE Transactions on Image Processing*, 16(8):2080–2095, 2007.

[5] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017.

[6] D. Verbin and T. Zickler. Field of Junctions: Extracting boundary structure at low SNR. In *ICCV*, 2021.

[7] N. Moeini, N. Kim, J. Romberg, and S. Fridovich-Keil. 3D Field of Junctions: A noise-robust, training-free structural prior for volumetric inverse problems. In *ECCV*, 2026. arXiv:2603.02149. Code: https://github.com/voilalab/3D-Field-of-Junctions.

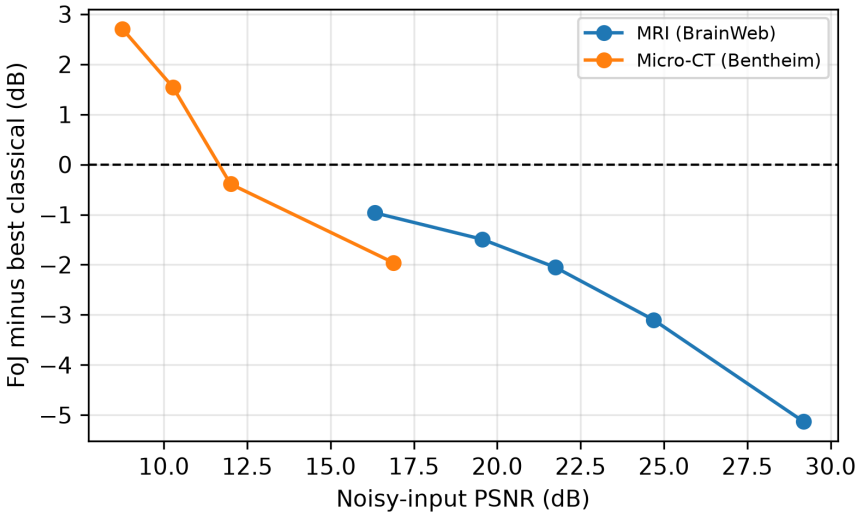


Fig. 5: FoJ PSNR margin over the best classical denoiser versus input PSNR. Above ≈11 dB input, classical filters are better; below it, FoJ is better on piecewise-constant volumes.

[8] S. Berg, H. Ott, D. Armstrong, A. Schwing, J. Humphrey, and R. Necker. High-resolution X-ray computed tomography images of Bentheim sandstone (1 MPa, 5.854 μm). Zenodo, https://doi.org/10.5281/zenodo.5548582, 2021.

[9] D. L. Collins, A. P. Zijdenbos, V. Kollokian, J. G. Sled, N. J. Kabani, C. J. Holmes, and A. C. Evans. Design and construction of a realistic digital brain phantom. *IEEE Transactions on Medical Imaging*, 17(3):463–468, 1998.

[10] R. Neumann, M. Andreeta, and E. Lucas-Oliveira. 11 sandstones: raw, filtered and segmented data. Digital Rocks Portal, https://doi.org/10.17612/F4H1-W124, 2020.